

Spectral Bound of Attention

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0. Notation (aligned with *Attention Is All You Need*). We follow the standard scaled dot-product attention notation. Let n denote the sequence length, and let d denote the model dimension. For a single head with key/query dimension d_k (so $h := d/d_k$ heads if multi-head is used), define

$$Q := XW_Q, \quad K := XW_K, \quad V := XW_V,$$

and the row-wise softmax attention

$$P(X) := \text{SM}\left(\frac{1}{\sqrt{d_k}} QK^\top\right).$$

Equivalently, letting

$$A := W_Q W_K^\top \in \mathbb{R}^{d \times d}, \quad S(X) := \frac{1}{\sqrt{d_k}} X A X^\top \in \mathbb{R}^{n \times n},$$

we have $P(X) = \text{SM}(S(X))$.

Throughout, $\|\cdot\|_2$ denotes the spectral norm for matrices and the Euclidean norm for vectors. For a linear operator $\mathcal{T}$ between matrix normed spaces, $\|\mathcal{T}\|_2$ denotes the induced operator norm.

1. Definition. We define our function $f(X)$ as the product of two matrix-valued functions $P(X)$ and $V(X)$:

$$f(X) = P(X) V(X).$$

Here $P(X)$ is the attention matrix of size $n \times n$, and

$$V(X) = XW_V$$

is the value matrix of size $n \times d_v$ (in the Transformer one often has $d_v = d_k$, but we do not need this here).

2. Perturbation. To find the Jacobian (Fréchet derivative), perturb X by a small matrix ΔX :

$$f(X + \Delta X) = P(X + \Delta X) V(X + \Delta X).$$

By first-order (Fréchet) expansion,

$$\begin{aligned} P(X + \Delta X) &\approx P(X) + (D_X P)[\Delta X] := P + \Delta P, \\ V(X + \Delta X) &\approx V(X) + (D_X V)[\Delta X] := V + \Delta V. \end{aligned}$$

3. Expand the product. Substitute into the product:

$$f(X + \Delta X) \approx (P + \Delta P)(V + \Delta V),$$

and distribute:

$$f(X + \Delta X) \approx PV + (\Delta P)V + P(\Delta V) + (\Delta P)(\Delta V).$$

The last term $(\Delta P)(\Delta V)$ is higher order: since P and V are Fréchet differentiable, $\|\Delta P\|_2 = O(\|\Delta X\|_2)$ and $\|\Delta V\|_2 = O(\|\Delta X\|_2)$, hence $\|(\Delta P)(\Delta V)\|_2 = O(\|\Delta X\|_2^2)$, so it vanishes faster as $\Delta X \rightarrow 0$ and is ignored in the derivative.

4. Isolate the linear part. Thus the first-order variation is

$$f(X + \Delta X) - f(X) \approx (\Delta P)V + P(\Delta V).$$

Replacing ΔP and ΔV by their Fréchet forms,

$$\Delta P = (D_X P)[\Delta X], \quad \Delta V = (D_X V)[\Delta X],$$

we obtain the Jacobian operator

$$\mathcal{J}(\Delta X) = (\textcolor{red}{D_X P})[\Delta X] \cdot \textcolor{red}{V} + \textcolor{blue}{P} \cdot (\textcolor{blue}{D_X V})[\Delta X].$$

Bounding the second term $\textcolor{blue}{P} \cdot (\textcolor{blue}{D_X V})[\Delta X]$. Since $(D_X V)[\Delta X] = \Delta X W_V$. Using the submultiplicativity of the spectral norm ($\|YZ\|_2 \leq \|Y\|_2 \|Z\|_2$):

$$\|P(\Delta X)W_V\|_2 \leq \|P\|_2 \|\Delta X\|_2 \|W_V\|_2. \quad (1)$$

Bounding the first term $(\textcolor{red}{D_X P})[\Delta X] \cdot \textcolor{red}{V}$. We define $\Delta S := D_X S(X)[\Delta X]$ as the variation of the score matrix $S = \frac{1}{\sqrt{d_k}} X A X^\top$.

The crucial identity $(\Delta P)\mathbf{1} = (D_S \text{SM}(S)[\Delta S])\mathbf{1} = \mathbf{0}$.

We claim that the rows of the softmax derivative ΔP sum to zero: $(\Delta P)\mathbf{1} = \mathbf{0}$.

Let $S : \mathbb{R}^{n \times d} \rightarrow \mathbb{R}^{n \times n}$ be a differentiable map and let

$$P(X) := \text{SM}(S(X)),$$

where $\text{SM} : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ denotes the row-wise softmax.

For any direction $\Delta X \in \mathbb{R}^{n \times d}$, define

$$\Delta P := D_X P(X)[\Delta X].$$

By the Fréchet chain rule,

$$\Delta P = D_X P(X)[\Delta X] = D_S \text{SM}(S(X)) [D_X S(X)[\Delta X]] = D_S \text{SM}(S) [\Delta S], \quad (2)$$

where we set

$$\Delta S := D_X S(X)[\Delta X].$$

Therefore to prove $(\Delta P)\mathbf{1} = \mathbf{0}$, it suffices to show that for any ΔS ,

$$(D_S \text{SM}(S)[\Delta S])\mathbf{1} = \mathbf{0}. \quad (3)$$

This follows from the fact that, for every S , the row-wise softmax produces a row-stochastic matrix:

$$\text{SM}(S)\mathbf{1} = \mathbf{1}.$$

Differentiating the identity $\text{SM}(S)\mathbf{1} = \mathbf{1}$ with respect to S in the direction ΔS yields

$$D_S(\text{SM}(S)\mathbf{1})[\Delta S] = \mathbf{0}.$$

By the proof below, we have that

$$D_S(\text{SM}(S)\mathbf{1})[\Delta S] = (D_S \text{SM}(S)[\Delta S])\mathbf{1}.$$

Combining the last two displays proves (3). Consequently, (3) holds for any ΔS , regardless of how ΔS is produced (in particular, even when $\Delta S = D_X S(X)[\Delta X]$ via (2)).

A rigorous proof of $D_S(\text{SM}(S)\mathbf{1})[\Delta S] = (D_S \text{SM}(S)[\Delta S])\mathbf{1}$

Let $\text{SM} : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ be the row-wise softmax map and define

$$F(S) := \text{SM}(S)\mathbf{1} \in \mathbb{R}^n.$$

We prove that for every direction $\Delta S \in \mathbb{R}^{n \times n}$,

$$DF(S)[\Delta S] = (D \text{SM}(S)[\Delta S])\mathbf{1}. \quad (2)$$

Step 1: Introduce the right-multiplication operator. Define the linear map

$$R_1 : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^n, \quad R_1(A) := A \cdot \mathbf{1}.$$

Then $F = R_1 \circ \text{SM}$, i.e.,

$$F(S) = R_1(\text{SM}(S)).$$

Lemma (boundedness of R_1). With the spectral norm $\|\cdot\|_2$ on matrices and the Euclidean norm $\|\cdot\|_2$ on vectors, R_1 is bounded:

$$\|R_1(A)\|_2 = \|A\mathbf{1}\|_2 \leq \|A\|_2 \|\mathbf{1}\|_2 \quad \forall A \in \mathbb{R}^{n \times n}.$$

Hence R_1 is continuous.

Step 2: Use the Fréchet definition (small-o) for SM. Assume SM is Fréchet differentiable at S . Then there exists a linear map $D \text{SM}(S) : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ such that as $\Delta S \rightarrow 0$,

$$\text{SM}(S + \Delta S) = \text{SM}(S) + D \text{SM}(S)[\Delta S] + r(\Delta S), \quad \frac{\|r(\Delta S)\|_2}{\|\Delta S\|_2} \rightarrow 0. \quad (3)$$

Step 3: Apply R_1 to the expansion and control the remainder. Apply R_1 to (3):

$$F(S + \Delta S) = F(S) + R_1(D \text{SM}(S)[\Delta S]) + R_1(r(\Delta S)).$$

Since $R_1(A) = A\mathbf{1}$, this is

$$F(S + \Delta S) = F(S) + (D \text{SM}(S)[\Delta S])\mathbf{1} + r(\Delta S)\mathbf{1}. \quad (4)$$

Using boundedness of R_1 ,

$$\frac{\|r(\Delta S)\mathbf{1}\|_2}{\|\Delta S\|_2} \leq \|\mathbf{1}\|_2 \frac{\|r(\Delta S)\|_2}{\|\Delta S\|_2} \longrightarrow 0 \quad (\Delta S \rightarrow 0),$$

so $r(\Delta S)\mathbf{1} = o(\|\Delta S\|_2)$.

Step 4: Identify the Fréchet derivative of F . From (4) and the previous estimate,

$$F(S + \Delta S) - F(S) - (D \text{SM}(S)[\Delta S])\mathbf{1} = o(\|\Delta S\|_2).$$

By the definition of the Fréchet derivative, this proves (2), i.e.,

$$D_S(\text{SM}(S)\mathbf{1})[\Delta S] = (D_S \text{SM}(S)[\Delta S])\mathbf{1}.$$

From $(\Delta P)\mathbf{1} = 0$ to $(\Delta P)\Pi Z = (\Delta P)Z$

Assume we have already established the *zero row-sum* property

$$(\Delta P)\mathbf{1} = \mathbf{0}.$$

Let

$$\Pi := I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$$

be the orthogonal projector onto $\mathbf{1}^\perp$. Then for any matrix (or vector) Z of compatible size,

$$(\Delta P)\Pi Z = (\Delta P)Z.$$

Starting from the left-hand side and substituting the definition of Π ,

$$(\Delta P)\Pi Z = (\Delta P)\left(I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)Z.$$

Using distributivity of matrix multiplication,

$$(\Delta P)\Pi Z = (\Delta P)IZ - \frac{1}{n}(\Delta P)\mathbf{1}\mathbf{1}^\top Z.$$

Since $(\Delta P)IZ = (\Delta P)Z$ and matrix multiplication is associative, we may regroup the second term as

$$(\Delta P)\Pi Z = (\Delta P)Z - \frac{1}{n}((\Delta P)\mathbf{1})\mathbf{1}^\top Z.$$

Finally, by $(\Delta P)\mathbf{1} = \mathbf{0}$, the second term vanishes:

$$(\Delta P)\Pi Z = (\Delta P)Z - \frac{1}{n}\mathbf{0}\mathbf{1}^\top Z = (\Delta P)Z.$$

Therefore, combining the results above, the first term $(D_X P)[\Delta X] \cdot V$ can be equivalently written as:

$$\Delta P V = (\Delta P)\Pi V = (\Delta P)\Pi X W_V,$$

whence

$$\|\Delta P V\|_2 \leq \|\Delta P\|_2 \|\Pi X\|_2 \|W_V\|_2.$$

Chain rule along a line via the Fréchet derivative

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be Fréchet differentiable at every point of the line $u(t) = s + t\Delta s$, and define

$$g(t) := f(u(t)) = f(s + t\Delta s).$$

Fix $t \in \mathbb{R}$. By the definition of the (one-dimensional) derivative,

$$g'(t) = \lim_{h \rightarrow 0} \frac{g(t+h) - g(t)}{h} = \lim_{h \rightarrow 0} \frac{f(u(t+h)) - f(u(t))}{h}.$$

Since $u(t+h) = s + (t+h)\Delta s = u(t) + h\Delta s$, this becomes

$$g'(t) = \lim_{h \rightarrow 0} \frac{f(u(t) + h\Delta s) - f(u(t))}{h}.$$

Because f is Fréchet differentiable at $u(t)$, there exists a bounded linear map $Df(u(t)) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and a remainder $r(v)$ such that, for $v \rightarrow 0$,

$$f(u(t) + v) = f(u(t)) + Df(u(t))[v] + r(v), \quad \frac{\|r(v)\|}{\|v\|} \rightarrow 0.$$

Apply this with $v = h\Delta s$:

$$f(u(t) + h\Delta s) - f(u(t)) = Df(u(t))[h\Delta s] + r(h\Delta s).$$

Divide by h and use linearity of $Df(u(t))$:

$$\frac{f(u(t) + h\Delta s) - f(u(t))}{h} = Df(u(t))[\Delta s] + \frac{r(h\Delta s)}{h}.$$

Moreover,

$$\left\| \frac{r(h\Delta s)}{h} \right\| = \frac{\|r(h\Delta s)\|}{\|h\Delta s\|} \|\Delta s\| \xrightarrow{h \rightarrow 0} 0,$$

hence the limit exists and equals $Df(u(t))[\Delta s]$. Therefore,

$$g'(t) = Df(u(t))[\Delta s] = Df(s + t\Delta s)[\Delta s].$$

Row-wise exponential normalization: exact increment and spectral bound. Let $\text{SM} : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ be the softmax function

$$(\text{SM}(S))_{i:} := \text{SM}(S_{i:}), \quad \text{SM}(s)_k := \frac{e^{s_k}}{\sum_{j=1}^n e^{s_j}}.$$

Fix $S \in \mathbb{R}^{n \times n}$ and a perturbation $\Delta S \in \mathbb{R}^{n \times n}$. Define

$$S(t) := S + t\Delta S, \quad P(t) := \text{SM}(S(t)), \quad p_i(t) := P(t)_{i:}.$$

Then for each row i we have the *exact* identity

$$\Delta P_{i:}^{\text{inc}} := P(1)_{i:} - P(0)_{i:} = \int_0^1 J(p_i(t)) \Delta S_{i:} dt, \quad J(p) := \text{Diag}(p) - p^\top p. \quad (5)$$

Derivation. Set $g_i(t) := p_i(t) = \text{SM}(S_{i:} + t\Delta S_{i:})$. Since SM is C^1 , g_i is differentiable and, by the chain rule in Fréchet form,

$$g'_i(t) = D \text{SM}(S_{i:} + t\Delta S_{i:})[\Delta S_{i:}].$$

Moreover, the Fréchet derivative of SM satisfies

$$D \text{SM}(u)[h] = J(\text{SM}(u))h.$$

Hence $g'_i(t) = \Delta S_{i:} J(p_i(t))$, and integrating from 0 to 1 yields (5).

Norm bounds. In the following section, $\|\cdot\|_2$ denotes the *Euclidean Norm*.

Let

$$\alpha := \max_{1 \leq i \leq n} \sup_{t \in [0,1]} \|J(p_i(t))\|_2.$$

Using $\|Ax\|_2 \leq \|A\|_2 \|x\|_2$ and (5),

$$\|\Delta P_{i:}^{\text{inc}}\|_2 \leq \int_0^1 \|J(p_i(t))\|_2 \|\Delta S_{i:}\|_2 dt \leq \alpha \|\Delta S_{i:}\|_2.$$

Since for any matrix Y , we have that $\|Y\|_F^2 = \sum_i \|Y_{i:}\|_2^2$, squaring and summing over i gives

$$\|\Delta P^{\text{inc}}\|_F \leq \alpha \|\Delta S\|_F.$$

Finally, using $\|M\|_2 \leq \|M\|_F$ and $\|M\|_F \leq \sqrt{\text{rk}(M)} \|M\|_2$,

$$\|\Delta P^{\text{inc}}\|_2 \leq \|\Delta P^{\text{inc}}\|_F \leq \alpha \|\Delta S\|_F \leq \alpha \sqrt{\text{rk}(\Delta S)} \|\Delta S\|_2.$$

Lemma (rank bound used above). With $\Delta S = \frac{1}{\sqrt{d_k}}((\Delta X)AX^\top + XA(\Delta X)^\top)$,

$$\text{rk}(\Delta S) \leq \text{rk}((\Delta X)AX^\top) + \text{rk}(XA(\Delta X)^\top) \leq \text{rk}(\Delta X) + \text{rk}(\Delta X) = 2 \text{rk}(\Delta X) \leq 2 \min\{n, d\} \leq 2d.$$

Therefore,

$$\|\Delta P^{\text{inc}}\|_2 \leq \alpha \sqrt{2d} \|\Delta S\|_2.$$

In particular, since $\sup_{p \in \Delta^{n-1}} \|J(p)\|_2 = \frac{1}{2}$, we have $\alpha \leq \frac{1}{2}$ and thus

$$\|\Delta P^{\text{inc}}\|_2 \leq \sqrt{\frac{d}{2}} \|\Delta S\|_2.$$

By our analysis on ΔP , we have that

$$\|\Delta P\|_2 \leq \sqrt{2d} \left(\max_i \sup_{t \in [0,1]} \|J(p_i(t))\|_2 \right) \|\Delta S\|_2, \quad \alpha := \max_i \sup_{t \in [0,1]} \|J(p_i(t))\|_2 \leq \frac{1}{2}.$$

Bound for ΔS under spectral norm

We differentiate $S = \frac{1}{\sqrt{d_k}} XAX^\top$:

$$\Delta S = \frac{1}{\sqrt{d_k}} \left((\Delta X)AX^\top + XA(\Delta X)^\top \right). \quad (6)$$

Bounding the spectral norm using triangle inequality and sub-multiplicativity:

$$\|\Delta S\|_2 \leq \frac{1}{\sqrt{d_k}} \left(\|(\Delta X)AX^\top\|_2 + \|XA(\Delta X)^\top\|_2 \right), \quad (7)$$

$$\leq \frac{1}{\sqrt{d_k}} \left(\|\Delta X\|_2 \|AX^\top\|_2 + \|XA\|_2 \|(\Delta X)^\top\|_2 \right). \quad (8)$$

Note that for the spectral norm, $\|Z\|_2 = \|Z^\top\|_2$. Thus $\|(\Delta X)^\top\|_2 = \|\Delta X\|_2$.

$$\|\Delta S\|_2 \leq \frac{1}{\sqrt{d_k}} \left(\|XA^\top\|_2 + \|XA\|_2 \right) \|\Delta X\|_2. \quad (9)$$

Combining all these terms together, we can finalize our bound for $\|\mathcal{J}\|_2$.

Indeed, for any ΔX ,

$$\|\mathcal{J}(\Delta X)\|_2 \leq \|W_V\|_2 \left(\|P\|_2 + \sqrt{2} \alpha \sqrt{\frac{d}{d_k}} \|\Pi X\|_2 \left(\|XA^\top\|_2 + \|XA\|_2 \right) \right) \|\Delta X\|_2,$$

hence by the definition of the induced operator norm,

$$\|\mathcal{J}\|_2 \leq \|W_V\|_2 \left(\|P\|_2 + \sqrt{2} \alpha \sqrt{\frac{d}{d_k}} \|\Pi X\|_2 \left(\|XA^\top\|_2 + \|XA\|_2 \right) \right),$$

where $h := d/d_k$ is the number of heads in the attention (to be an integer).